On Transferring Surgical Policies from Simulation to Reality: Perception, Calibration, and Safety Control

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Although surgical simulation provides an effective environment for policy development, transferring learned behaviors to physical robots remains challenging. We present a modular framework for deployment on a physical da Vinci Research Kit (dVRK), using reinforcement-learning (RL) suturing policies trained in SurgicAI, a simulation-based surgical policy learning platform, as a case study. Directly transferring simulation-trained policies to the physical robot using only observation–action conversion, without additional deployment modules, resulted in 0% task success in the tested configurations. Transferring these policies to the physical dVRK requires addressing gaps in stereo perception, camera–robot calibration, geometric alignment, safety-constrained Cartesian control, and task verification. This work introduces dedicated deployment modules to mitigate these gaps. Following exploratory refinement, an operator visually confirmed pre-insertion alignment of the retained needle in 29 of 30 placing trials (96.7%). Heavy needle occlusion occurred in 11 trials. After the Reach policy brings the instrument to the needle, grasping uses no learned policy: a perception-guided geometric procedure aligns the jaws, descends, closes, and lifts, and it retained the needle in 40 of 50 final trials (80%). Separating policy-specific observation–action mappings from physical deployment modules provides a modular framework for integrating other compatible surgical policies. These results demonstrate the feasibility of transferring simulation-trained surgical policies to the physical dVRK by addressing real-world perception, calibration, and control challenges. The source code is publicly available at GitHub.

Keywords: surgical robotics; sim-to-real transfer; reinforcement learning; dVRK; needle grasping; stereo vision; robot calibration.

1. Introduction

Robotic surgical suturing involves precise needle manipulation to join tissues, making it a challenging task for automation. Simulation enables surgical robot policies to learn these precise, contact-rich behaviors without the cost of repeated physical trials. Surgical simulation environments provide reproducible task configurations and direct access to robot and object states, supporting scalable training for needle manipulation [1–3]. However, a policy that succeeds in simulation cannot be deployed on the physical system, such as da Vinci Research Kit (dVRK) [4], simply by forwarding its actions to the robot. The observations, coordinate conventions, instrument geometry, and task-completion conditions available during training must be connected to corresponding physical measurements and execution procedures.

Needle manipulation couples perception, calibration, control, and verification. Reflections, occlusion, and depth uncertainty complicate visual pose estimation; camera–robot registration must reconcile reported poses with physical jaw geometry [5, 6]. Actions require consistent units, frames, and scaling, while execution must respect joint, workspace, and contact constraints. Secure retention also requires physical verification. Visual adaptation alone [7] cannot resolve these deployment gaps.

We address these deployment gaps through a modular framework that connects simulation-trained suturing policies to the physical dVRK. The framework combines stereo-based needle and suturing-site perception, camera–robot calibration, instrument-geometry compensation, policy-specific observation and action mappings, safety-filtered Cartesian execution, and task-level verification. Its central design separates the learned controller from the physical deployment infrastructure: the observation–action mapping expresses measured task states in the policy's observation convention and converts policy outputs into appropriately scaled Cartesian commands, while shared modules handle sensing, calibration, execution constraints, and verification. Other compatible policies require interface adaptation and physical validation.

The Surgical Robotics Challenge [1] provides a standardized, dVRK-compatible simulation environment for developing robotic suturing policies. Built upon this environment, SurgicAI [3] achieved an 88% success rate on the simulated Approaching–Placing task sequence, making it a suitable case study for investigating sim-to-real policy transfer.

To evaluate its applicability to physical robotic manipulation, we deploy the simulation-trained SurgicAI policies for Reach and Place on the physical dVRK using a Cartesian command interface compatible with the Collaborative Robotics Toolkit (CRTK) [8]. Our initial deployment, which converted physical observations into policy inputs and policy outputs into robot motion commands without additional deployment modules, achieved 0% task success in the tested configurations. This result highlights the need to address the discrepancies between simulation and

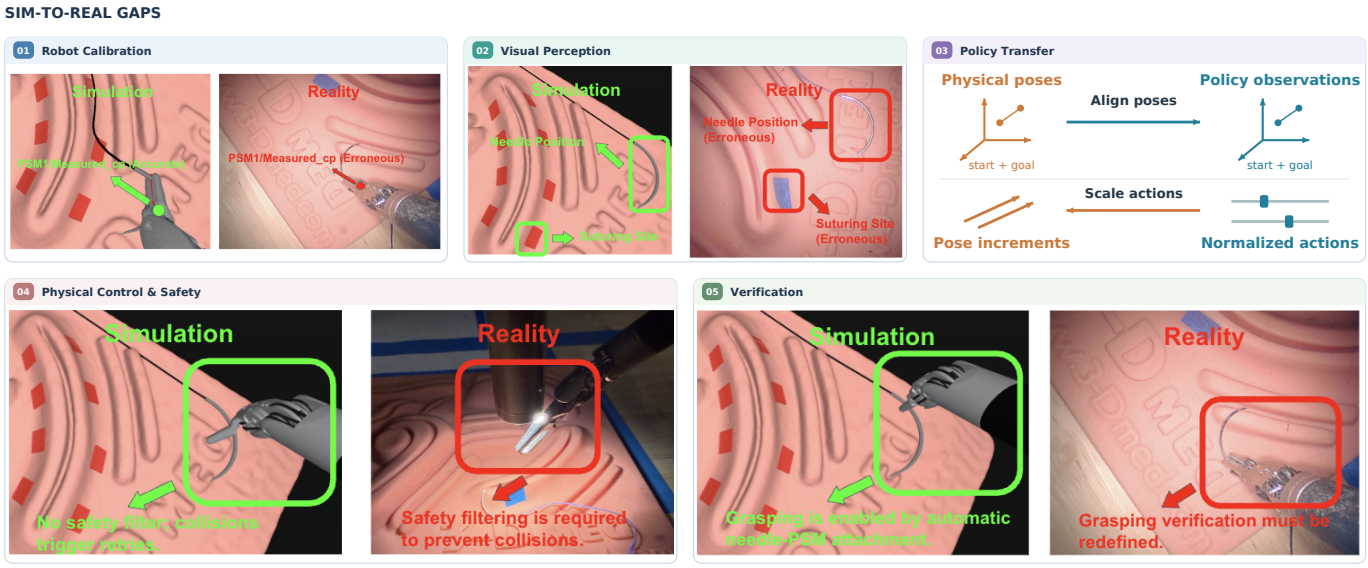

Fig. 1. Sim-to-real gaps in dVRK deployment: (1) robot calibration, (2) visual perception, (3) policy observation and action transfer, (4) physical control and safety, and (5) physical grasp verification.

physical execution. Because physical needle and start poses also exceeded the policy's training range, we retrained the Reach policy in SurgicAI over an expanded range (Sec. 3.4). Grasping is not executed by a learned policy. In simulation the needle is attached to the jaws automatically, so no grasp behavior is available to transfer; on the robot, a perception-guided geometric procedure aligns the jaws with the needle, descends, closes, and lifts. In separate-stage physical evaluations, this procedure, executed after the Reach policy, retained the needle in 40 of 50 grasping trials (80%), and the Place policy with the deployment modules met the operator-assessed placing criterion in 29 of 30 trials (96.7%).

The main contributions of this work are:

- A modular framework for bridging perception, calibration, action-interface, execution, and verification gaps between simulated suturing policies and the physical dVRK.
- A separation of policy-specific observation–action mappings from shared physical deployment modules, providing an interface-based route to integrating additional compatible suturing policies.
- A physical deployment case study in an extended workspace, with SurgicAI RL policies for Reach and Place and a geometric procedure without a learned policy for Grasp, documenting geometric adjustments and the perceptual, geometric, and contact-related limitations observed in grasping and placing experiments.

2. Related Work

Surgical robotics simulation frameworks provide task-specific environments for developing and evaluating autonomous manipulation policies. Sagitov et al. [9] extended Gazebo to simulate surgical tissue and suture interactions, while SurRoL [10] provides a dVRK-compatible reinforcement-learning platform for surgical manipulation. ORBIT-Surgical [11] leverages NVIDIA Isaac Sim for scalable surgical skill learning, and the AMBF-based Surgical Robotics Challenge [1] provides a dedicated environment for robotic suturing. Although domain randomization can reduce visual discrepancies [7], physical deployment additionally requires calibrated perception, consistent action interpretation, and reliable execution. We investigate these challenges through SurgicAI deployment on the physical dVRK.

Within surgical robotics, human-in-the-loop simulation facilitates demonstration collection [2], and SurgicAI [3] supports hierarchical learning of suturing subtasks using RL and imitation learning. Ou and Tavakoli [12] demonstrated physical transfer of an RL policy for tissue-point manipulation. Other approaches include needle pick-and-place without expert demonstrations [13] and imitation-learned da Vinci manipulation using relative actions to address imperfect kinematics [14]. Model-based autonomous anastomosis [15] and hierarchical language-conditioned imitation learning [16] further demonstrate progress toward surgical autonomy. Our focus is the sensing, control, and verification infrastructure needed to connect simulation-trained policies to physical needle manipulation.

State-based policies require physical estimates of object geometry supplied directly in simulation. YOLO [17], FoundationStereo [18], and FoundationPose [19] provide image detection, dense depth, and rigid-object pose estimation, respectively. Thin reflective needles and instrument occlusion additionally require task-specific reconstruction

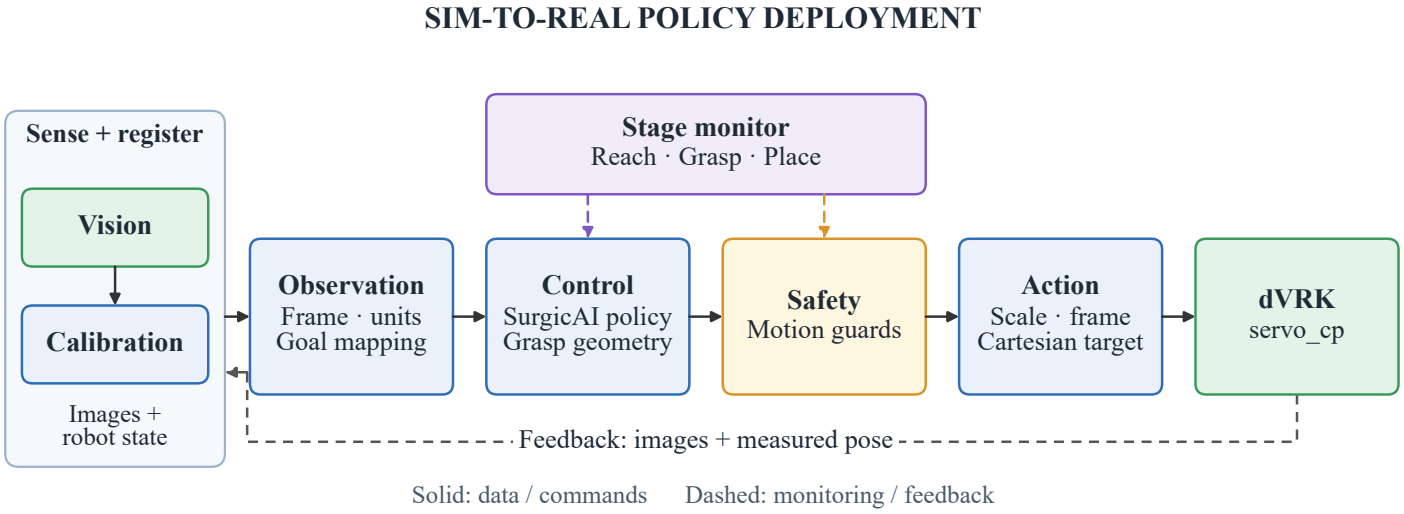

Fig. 2. Overview of the modular sim-to-real surgical policy deployment framework. The observation mapping encodes the measured state and goal for the policy. The action mapping converts policy outputs into Cartesian commands. The execution controller checks commands before execution and monitors tracking afterward. Updated visual state estimates and robot measurements provide feedback.

and reliable camera-tool registration.

STITCH 2.0 [20] integrates needle pose estimation, filtering, and manipulation primitives to improve robotic suturing. Separately, Joglekar et al. [21] propose reliability-driven surgical thread reconstruction and grasp planning under noisy visual observations. These studies motivate our separation of policy interfaces from calibrated perception, safety-filtered execution, and stage verification.

3. Methods

3.1. *Sim-to-Real Gaps and System Overview*

Deploying simulation-trained surgical policies on the physical dVRK requires addressing five interconnected sim-to-real gaps, as illustrated in Fig. 1. Robot calibration must reconcile camera observations with physical instrument geometry, while visual perception must replace simulator-provided needle and suturing-site states with estimates from stereo images. Policy transfer requires adapting simulation-specific observation conventions and normalized actions to physical robot coordinates and motion scales. Physical control and safety must account for workspace limitations, collision risks, and uncertain contact. Finally, task verification must replace simplified simulation-defined completion conditions with physical evidence of successful needle acquisition, retention, and placement.

To address these gaps, we introduce the modular deployment framework illustrated in Fig. 2. The pipeline begins with stereo perception and camera–robot calibration, which reconstruct the physical task state and express it in consistent reference frames. Policy-specific observation mappings transform these measurements and task goals into the representations expected by the SurgicAI

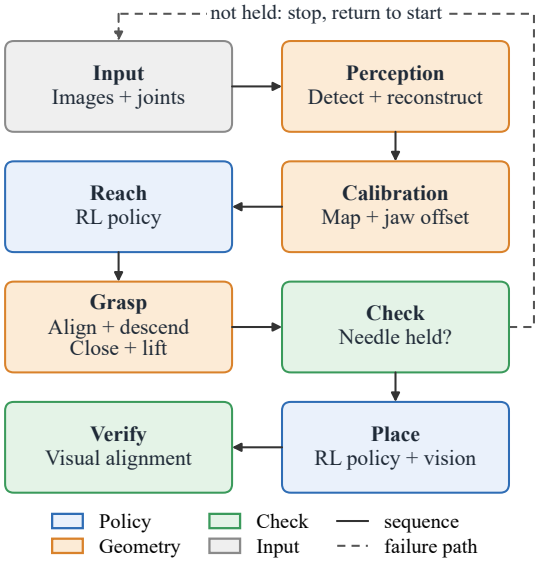

Fig. 3. Reach, Grasp, and Place in the deployment workflow. SurgicAI RL policies execute Reach and Place; Grasp is a geometric procedure without a learned policy. Perception and registration supply task-state estimates, and geometric alignment, motion constraints, and task verification support execution. The physical experiments evaluate each stage separately.

RL policies [1, 3]. The action mapping scales and transforms the normalized policy motion outputs into Cartesian increments. The execution controller checks candidate commands from the action mapping and geometric alignment against the stage-dependent limits before sending them to the dVRK through the Collaborative Robotics Toolkit (CRTK)-compatible `servo_cp` interface. The controller monitors tracking using measured robot poses. Up-

dated state estimates and robot measurements provide feedback for subsequent policy execution.

Figure 3 illustrates how Reach, Grasp, and Place fit into the manipulation workflow. In Reach, stereo perception and camera–robot registration provide an estimate of the needle grasping region, and the Reach policy generates approach motions toward this region. Grasp uses no learned policy: geometric alignment positions the jaws from the estimated needle pose, followed by a scripted descent, closure, and lift. In Place, the policy generates alignment motions from estimates of the entry site and the retained needle pose. During temporary occlusion, the system predicts the needle pose from the measured gripper pose and the last reliable needle–gripper transform (Sec. 3.3). We evaluated each stage separately on the physical dVRK (Sec. 4).

3.2. *Robot Calibration*

Errors in instrument references and measured poses can misalign the needle and gripper even when the visual target is correctly estimated. We therefore combine camera–robot registration, instrument-geometry compensation, and local position correction.

As illustrated in Fig. 4, C denotes the camera frame, E the ECM mechanical tip, and M the PSM1 reported control point (`measured_cp`). The distal jaw tips J_1 and J_2 define the gripper center J_c at their midpoint. Calibration uses $E \leftrightarrow M$ and $C \leftrightarrow J_c$ correspondences without direct $C \leftrightarrow E$ calibration.

To compensate for residual errors, we fit a local affine mapping from robot poses and stereo-tracked jaw observations. The observed gripper center is

$$ {}^{C}\mathbf{x}_{J_c}^{\text{obs}} = \frac{1}{2}\left({}^{C}\mathbf{x}_{J_1} + {}^{C}\mathbf{x}_{J_2}\right). \tag{1} $$

Its predicted camera-frame position is

$$ \begin{aligned} {}^{C}\hat{\mathbf{x}}_{J_c}(p,R) &= A(p+Rd)+b+\beta, \\ d &= d_{\text{TCP}} + g_{\text{aim}}, \end{aligned} \tag{2} $$

where (p, R) is the measured PSM1 pose, (A, b) defines the affine mapping, d is the effective control-point-to-gripper offset, and β compensates for the tracked-feature reference. This mapping supports local needle–gripper alignment but does not guarantee global contact accuracy.

A Bernstein-polynomial correction for position-dependent residual errors (Fig. 5) was also investigated; it does not account for orientation-dependent errors, so the deployed system uses the local affine mapping.

3.3. *Visual Perception*

Unlike simulation, where needle and target poses are directly available, physical deployment estimates them from stereo images affected by reflections, occlusion, and depth uncertainty. We use FoundationPose [19] for needle pose estimation and FoundationStereo [18] for depth recovery.

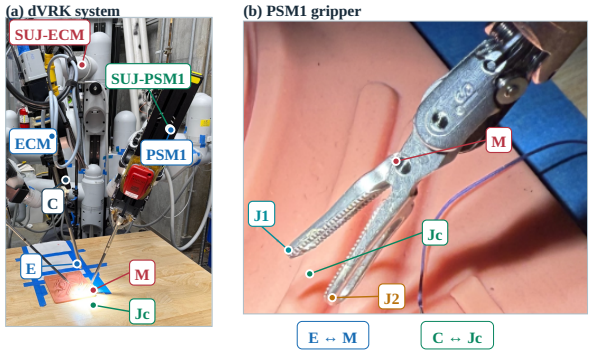

Fig. 4. Camera–robot relationships and instrument reference points used for grasp alignment.

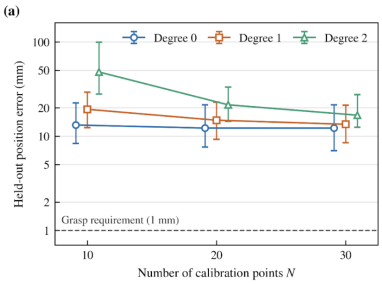

Fig. 5. Bernstein-polynomial residual correction investigated during preliminary robot calibration.

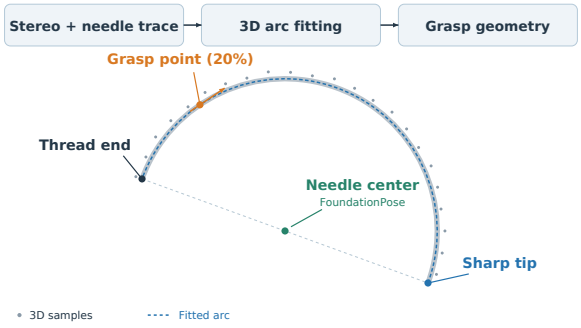

Fig. 6. Reconstructed needle arc, endpoints, grasp point at 20% of arc length from the thread-attached end, and FoundationPose-estimated needle center.

For grasp localization, the detected needle centerline is projected onto a reconstructed quadratic phantom surface and fitted to a planar circular arc, assuming contact with the supporting surface (Fig. 6). The grasp point is selected at 20% of the arc length from the thread-attached end, with the local tangent determining jaw alignment:

$$ \theta_g = \theta_{\text{end}} + 0.20(\theta_{\text{tip}} - \theta_{\text{end}}), \quad g = p(\theta_g). \tag{3} $$

Here, $p(\theta)$ parameterizes the fitted arc, and θ_{end} and θ_{tip} are its unwrapped endpoint angles. Geometric consistency checks assess reconstruction quality before grasping.

For placing, Lab color segmentation identifies the blue

suturing-site marker, and stereo depth estimates its 3D position. During temporary needle occlusion, the last reliable needle–gripper relationship is propagated using measured robot motion:

$$ {}^{J}T_{O}(t_0) = \left({}^{P}T_{J}(t_0)\right)^{-1} {}^{P}T_{O}(t_0), \tag{4} $$

$$ {}^{P}\widehat{T}_{O}(t) = {}^{P}T_{J}(t)\, {}^{J}T_{O}(t_0). \tag{5} $$

This assumes the needle remains fixed relative to the gripper. Unreliable observations do not update the stored relative pose.

3.4. *Policy Transfer*

SurgicAI policies use simulation-specific states and produce normalized actions. Policy-specific observation–action mappings connect them to physical coordinates and Cartesian commands without modifying the learned controllers.

The observation mapping transforms calibrated robot states and task goals into policy-compatible coordinates while preserving relative geometry. The action mapping scales normalized outputs using training scales and stage-dependent gains, then converts them into Cartesian increments in the robot reference frame. For the policy-driven subtasks $s \in \{\text{Reach}, \text{Place}\}$,

$$ \begin{aligned} o_t^{(s)} &= \mathcal{O}_s(z_t, g_s), \\ a_t^{(s)} &= \pi_s(o_t^{(s)}), \\ \Delta T_t^{(s)} &= \mathcal{A}_s(a_t^{(s)}; z_t, g_s), \end{aligned} \tag{6} $$

where z_t is the measured robot and visual state, g_s is the subtask goal, π_s is the SurgicAI RL policy, and $\Delta T_t^{(s)}$ is a Cartesian increment in the physical robot frame. Task-goal adaptation and geometric alignment corrections extend the effective operating range without expanding the robot's mechanical workspace. Grasp lies outside this mapping: its commands are computed directly from the estimated needle geometry and pass through the same execution checks (Sec. 3.5).

The mappings cannot compensate for states that the policy never encountered during training. SurgicAI randomizes the needle and instrument start poses only within a limited range, whereas physical needle placements across the pad and slopes and session-dependent start poses often fall outside it, where the policy output is unreliable. We therefore retrained the Reach policy in SurgicAI over an expanded randomization range of needle and start poses, using the same training pipeline; the mappings apply to the retrained policy unchanged.

3.5. *Physical Control and Safety*

Physical execution introduces contact and workspace constraints only partially represented in simulation, so the controller checks requested Cartesian targets and monitors measured instrument poses (Fig. 7).

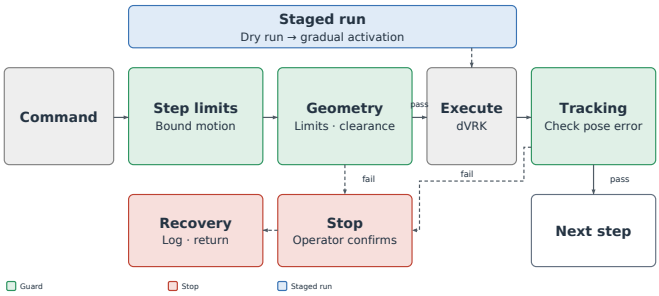

Fig. 7. Stage-dependent motion limits, command filtering, and post-command execution checks.

Operating at 10 Hz, the controller executes policy actions and geometric alignment corrections through the CRTK-compatible `servo_cp` interface. Free-space increments are limited to 5 mm and 5° per step, decreasing to 2 mm and 3° within 20 mm of the needle or phantom. Smaller increments apply during hover, descent, lifting, and fine alignment. Joint-limit margins, a local ±40 mm workspace, and instrument clearance above the supporting surface are also enforced.

Targets violating these constraints are rejected without clipping. The requested target is compared with the controller setpoint to detect inverse-kinematics clipping and with the measured pose to detect tracking errors. Execution stops when the measured pose differs from the requested target by more than 2 mm or 2°.

3.6. *Task Verification*

Simulation-defined completion conditions, including automatic needle–instrument attachment, neither provide a transferable grasp behavior nor guarantee physical needle acquisition or retention. Similarly, reaching a target pose does not confirm successful placement. We therefore verify grasping through visual needle displacement after closure and lifting, assessed by the symmetric Chamfer distance between pre- and post-lift needle observations, followed by retention during a two-second hold; movement alone does not confirm secure retention. For placing, an operator visually assesses final alignment of the retained needle after the robot settles, using the criterion in Sec. 4.2.

4. Case Study: Separate-Stage Reach–Grasp–Place Evaluation

4.1. *Experimental Design*

We evaluate three stages on the physical dVRK: Reach and Place, executed by the SurgicAI RL policies (with Reach retrained over the expanded range of Sec. 3.4), and Grasp, executed by a geometric procedure without a learned policy. Workspace adaptation maps policy commands from the simulated task workspace to the corresponding physical workspace. Geometric deployment adjustments use the

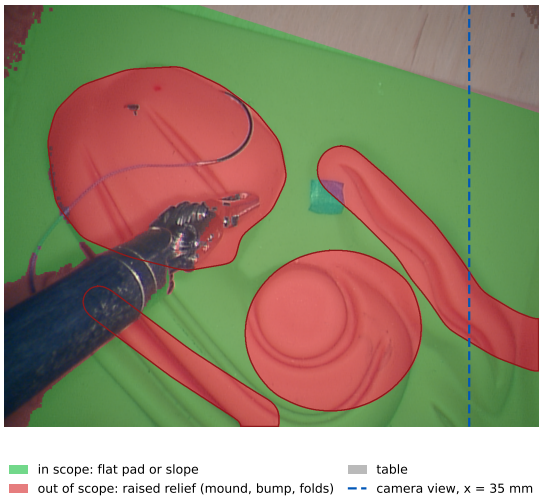

Fig. 8. Grasp scope on the phantom. Green indicates flat or sloped surfaces within scope, and red indicates raised mounds, bumps, and folds outside it. Right of the blue dashed line, only one camera has visibility, precluding stereo-guided grasping.

estimated needle pose to refine the instrument's position and orientation during alignment. We first conducted exploratory trials to identify execution problems, refined the deployment procedure based on those observations, and then evaluated the resulting procedure in a separate set of final trials. Reach and Place outcomes reflect the combined performance of the policy and the deployment modules; Grasp outcomes reflect the geometric procedure executed after Reach.

Grasping exploration comprised 30 trials across three setups. A separate 20-repeat Reach series used a fixed start and needle pose. Final acquisition comprised 50 evaluated trials over five calibrated sessions. In each trial, the SurgicAI Reach policy first moved the instrument to the pre-grasp pose at the estimated grasp point, and the geometric grasp procedure (jaw alignment to the local needle tangent, bounded descent, closure, lift, and hold) then executed acquisition without a learned policy. Reach was also evaluated in its separate repeat series; the acquisition outcomes characterize the Grasp stage. The evaluated grasping scope covers the flat pad and slopes; raised relief, including bumps, mounds, and folds, is outside this scope (Fig. 8).

Placing used ten exploratory trials per setting across three settings, followed by ten final trials per setting after refinement (Sec. 6.2). The final placing evaluation comprises 30 trials. Grasping and placing are evaluated separately: grasping with the geometric procedure, and placing with the SurgicAI Place policy and its deployment adjustments. A complete end-to-end Reach–Grasp–Place rollout, insertion, release, and complete suturing remain outside the present evaluation.

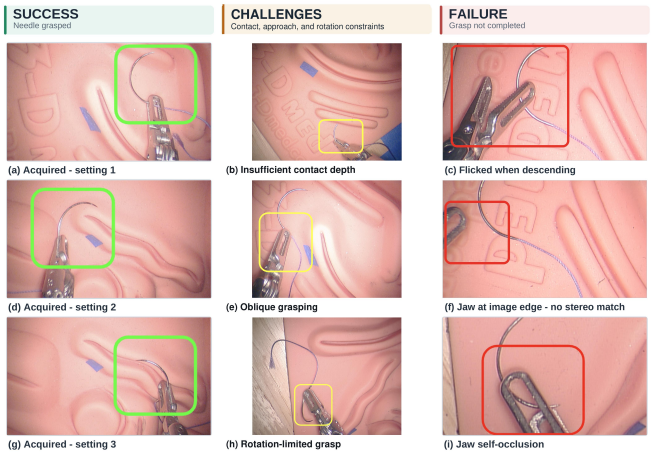

Fig. 9. Representative grasping outcomes.

4.2. *Evaluation Criteria*

A Reach trial was scored as successful when the measured instrument pose was within 5 mm and 15° of the approach goal. Each trial allowed up to 200 policy actions. The execution limits and tracking checks in Sec. 3.5 applied throughout the trial.

Exploratory grasps are classified as success (usable retention), semi-success (uncertain or unsuitable acquisition), or failure (no usable grasp). Final success requires retention through lifting and the two-second hold. Failures are assigned one cause: inaccurate needle fit, joint limit, phantom contact, needle displacement, or jaw self-occlusion.

An operator visually assessed the final needle configuration after the robot settled. A placing trial was scored as successful when the needle remained held by the gripper and the operator confirmed that its position and orientation at the intended entry site were suitable for insertion. Trials with unconfirmed final alignment were counted as failures. Tracking challenges were distinguished from terminal failures due to lost confirmation, rotational limits, or positional unreachability.

5. Results

5.1. *Reach and Needle Grasping Outcomes*

The Reach policy met its stopping criterion in all 20 fixed-configuration trials (100%). Representative grasping outcomes (Fig. 9) show successful retention (a,d,g), insufficient contact depth, oblique acquisition, and rotation-limited retention (b,e,h), and failures from needle displacement, missing stereo correspondence, and jaw self-occlusion (c,f,i).

Following the SurgicAI Reach policy, the geometric grasp procedure retained the needle in 40/50 trials (80%; Fig. 10). The ten failures included two inaccurate arc fits (trials 13,29), three approach failures (joint limit, phantom contact, or needle displacement), and five jaw self-occlusions. Retention was 23/30 on flat pads (≤ 1 mm) and

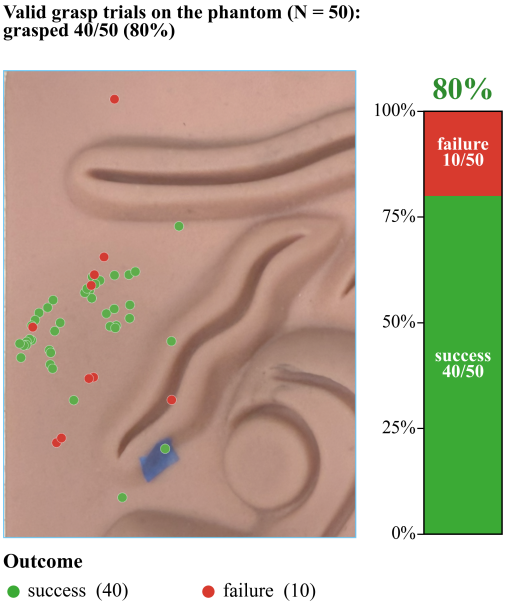

Fig. 10. Geometric grasp procedure: 50 evaluated grasp points, with 40 successful (green) and 10 unsuccessful (red).

17/20 on slopes (1–6 mm). All self-occlusions occurred on flat pads, while approach failures occurred on slopes at 4.2–5.2 mm. Among 47 trials with recorded closure roll, retention was 12/17 at negative roll and 28/30 at non-negative roll; all self-occlusions occurred between −68° and −32°, although successful grasps reached −85°. These results reflect separate-stage Grasp evaluation; the learned policy acts only in the preceding Reach.

5.2. *Needle Placing Outcomes*

Exploratory placing (Fig. 11) demonstrated successful alignment (a–c), occlusion and orientation ambiguity (d,e), and failures involving visibility, rotation, and reachability (f–h). Workspace maps (Fig. 12(a)) distinguish stereo-visible reachable regions (blue), visible regions beyond joint limits (orange), and single-camera regions (gray). Settings 1–3 covered 28.5, 28.5, and 19.8 cm^2, with reachable fractions of 94%, 100%, and 90%, respectively. Circles indicate grasp locations and stars indicate target sites.

Using the visual assessment criterion in Sec. 4.2, 29/30 final placing trials succeeded (96.7%), with 10/10, 10/10, and 9/10 successes across Settings 1–3 (Fig. 12(b)). Heavy occlusion occurred in 11/30 trials (36.7%). The single failure, in Setting 3, occurred within the reachable region (filled circle): post-grasp occlusion prevented needle–gripper initialization, and unsuccessful view-seeking moved PSM1 outside the camera view, leaving alignment unverified.

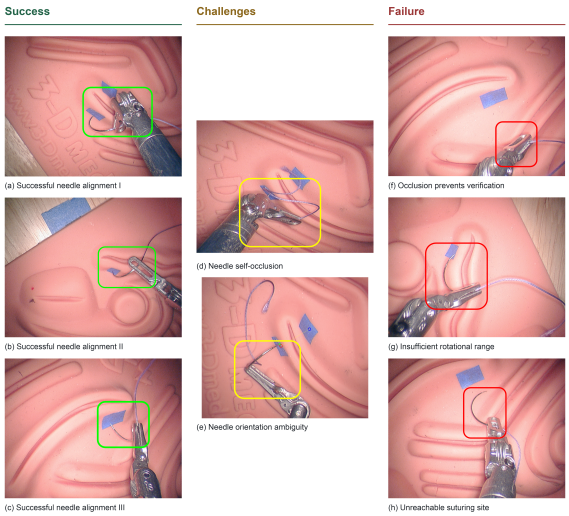

Fig. 11. Exploratory placing: (a–c) success; (d,e) occlusion and orientation ambiguity; (f–h) visibility, rotation, and reachability failures.

6. Discussion

6.1. *Needle Grasping: Refinements Informed by Exploration*

Jaw error, premature contact, and self-occlusion motivated setting-specific calibration, kinematic fallback, bounded descent, and effort monitoring after jaw alignment. Aim and reference-bias corrections are re-estimated for each setting, but calibration residuals indicate geometric consistency rather than contact accuracy. The 1 mm stereo jaw-depth criterion is an orientation heuristic, and uncalibrated effort cannot confirm capture. Post-lift retention is therefore required, while the stationary/moved/undetermined image check remains unvalidated on hardware.

Approach clearance and remaining wrist range further constrain acquisition, potentially requiring repositioning or alternative grasps. Five of the ten final failures involved jaw self-occlusion at negative roll, compared with none in 30 non-negative-roll trials. A fixed endoscope may obscure one jaw blade behind the other, requiring reliance on the roll model (Sec. 3.2). Its approximately 1 mm error approaches the needle-wire diameter, and checks using the same estimate cannot independently detect aim error.

However, 12/17 negative-roll trials succeeded, including at the largest angles, suggesting that pose-dependent prediction error may matter more than roll magnitude alone. Because roll was coupled to needle placement, this association is not causal evidence. Independent blade views or grasp-point selection favoring visible jaw configurations could test this mechanism. Matched ablations are also needed to separate the contributions of alignment, descent, and closure.

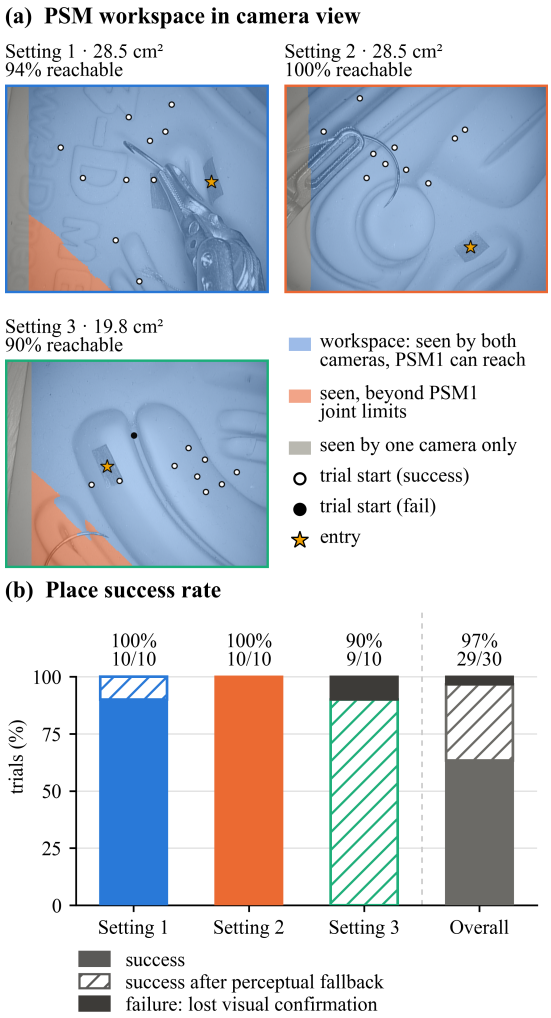

Fig. 12. Final placing evaluation. (a) Workspace and trial locations; open and filled circles mark grasp locations of successful and unsuccessful trials, and stars mark entry targets. (b) Outcomes by setting under the operator's visual assessment criterion; hatching indicates perceptual fallback, and the dark segment unconfirmed alignment.

6.2. *Needle Placing: Refinements and Remaining Failure Modes*

Occlusion and orientation ambiguity motivated confidence-triggered needle-pose propagation using the last reliable needle–gripper transform, assuming the needle remains fixed relative to the gripper. Final alignment is assessed visually (Sec. 4.2). The Place policy also requires accurate entry-marker and needle localization. Workspace maps (Fig. 12(a)) characterize positional accessibility: Settings 1 and 2 have equal areas but different reachable fractions, while Setting 3 is smaller. These maps do not guarantee visibility, rotational freedom, or collision-free trajectories.

The failure within the reachable region demonstrates that accessibility does not ensure visibility for needle-pose initialization. Gripper motion can propagate an initialized pose but cannot recover an unknown initial needle orientation. This motivates a pose-initialization check at the Grasp–Place handoff and bounded view-seeking that stops execution if no reliable view is recovered before PSM1 leaves the camera view. These proposed safeguards remain unvalidated. Matched comparisons are needed to isolate the effects of deployment refinements and workspace size.

6.3. *Implications and Limitations*

The results demonstrate physical execution of SurgicAI Reach and Place policies within an extended workspace, supported by calibrated perception, target localization, and execution constraints, together with a geometric Grasp procedure that uses no learned policy. However, no geometry-only baseline or matched ablations isolate the contribution of the Reach and Place policies from these deployment adjustments, and the grasping result says nothing about a learned grasp policy. Transferring additional policies would require adapting observations, action frames, scaling, and timing, retraining where the physical workspace exceeds the training range, and physical validation.

Evaluation remains stage-specific, without an end-to-end Reach–Grasp–Place success rate. Grasping was evaluated across five calibrated sessions on flat pads and slopes, excluding raised tissue features such as bumps and folds. Placing included 30 trials across three settings, with success assessed visually rather than through independent geometric measurements. Future work should address full-sequence evaluation, matched ablations, independent jaw views, and broader tissue conditions. Complete autonomous suturing additionally requires needle passage and thread handling.

7. Conclusion

We transferred SurgicAI RL policies for Reach and Place to the physical dVRK with workspace adaptation, calibrated perception, geometric alignment, safety-filtered execution, and verification, and performed Grasp with a geometric procedure without a learned policy. The Place policy generated motions to align the retained needle at the entry site. An operator visually confirmed pre-insertion alignment in 29 of 30 placing trials (96.7%). Heavy occlusion occurred in 11 trials. In the only unsuccessful trial, immediate post-grasp occlusion prevented initialization of the needle pose relative to the gripper. The geometric grasp procedure retained the needle in 40 of 50 trials (80%); five of ten failures involved negative-roll jaw self-occlusion. Component ablations, broader evaluation, and complete suturing remain future work.

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